CLINT MATHEWS

Senior Backend Engineer - Distributed Systems

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PROFESSIONAL SUMMARY

Lead Backend Software Engineer with 7+ years of experience building high-performance distributed systems for enterprise-scale platforms, specializing in microservices, Go, and reliability at scale.

TECHNICAL SKILLS

Backend and Messaging: NestJS, FastAPI, .NET Core, gRPC, Kafka, BullMQ, Redis, Celery
Languages: Go, Python, TypeScript, JavaScript, C#
Architecture: Microservices, event-driven architecture, circuit breakers, bulkhead isolation, backpressure, distributed locking, OCPP, RFC authorship
Cloud and Observability: AWS, GCP, Docker, GitHub Actions, CI/CD, Datadog, Prometheus, Grafana
AI Systems: Ollama, OpenRouter, Langfuse, Instructor, Pydantic, vectorized NumPy and SciPy
Data: PostgreSQL, MongoDB, MySQL, Redis, etcd

WORK EXPERIENCE

Ford Motor Company Aug 2022 - Present
Technical Lead (Senior Software Engineer) Remote, India

- Architected a high-throughput Kafka consumer in NestJS and TypeScript that averages 6M+ production OCPP charger messages daily, with higher peak volumes.
- Orchestrated end-to-end delivery of scalable backend services for a monolith-to-microservices transition, coordinating design, development, and deployment.
- Authored 3 foundational RFCs defining technical strategy for OCPP 2.x adoption and core platform services.
- Consolidated more than 2,500 depot-specific OCPP gateway instances into 6 shared pods through a phased three-month transition in 2025, initially migrating 8,200 chargers and saving approximately $9,000 per month primarily in compute costs.
- Led a 5-person charging engineering team from June 2025, directly owning 4 services and sharing ownership of 12 additional charging and utility services with assigned service owners.
- Drove development of a global OCPP simulator modeling 2,000+ chargers to validate gateway performance and reliability before production.
- Reduced Kafka consumer lag by improving downstream service performance and returning HTTP 201 acceptance before completing nonblocking asynchronous processing.
- Addressed recurring downstream service failures by decomposing a large monolith and introducing event-driven processing patterns to improve performance and isolation.
- Enhanced an explicit OCPP charger simulator to reproduce real charger behavior, improving connection and reconnection handling and expanding supported OCPP messages for targeted test flows.
- Translated Product feature requests into scoped technical requirements and coordinated engineering and hardware teams to plan and implement integration work end to end.

Experion Technologies Aug 2019 - Jul 2022
Software Engineer Hybrid, Kochi, India

- Directed technical delivery across 4 enterprise projects spanning frontend and backend systems while mentoring teams of up to 10 developers.
- Built ScanCo Connect to ingest data from an older on-premises system through AWS SNS and SQS fan-out, persist tenant-specific events, and display event-driven synchronization status in the UI.
- Delivered React frontend work for ARIA, a B2B SaaS application used to sell and consume products in bulk.
- Managed delivery across the full software development lifecycle, from requirements gathering through production deployment for enterprise clients.

PROJECTS

ChaosBox (GitHub | Medium | Series) *Go, PostgreSQL, Kubernetes, Minikube, Docker, Python*
Built a Go and PostgreSQL store service on a resource-constrained Minikube deployment, with transactional order creation, Testcontainers end-to-end coverage, and a Python load harness that validates complete user journeys.

Fencelock *Go, etcd, Redis, PostgreSQL, testcontainers-go*
Built a Go distributed-lock library where etcd issues monotonic fencing tokens and protected resources reject stale writes after a lock holder pauses beyond its lease TTL.

PhotonicOps *Go, Python, gRPC, Protobuf, Ollama, Docker*
Built an air-gapped Go and Python telemetry pipeline with mTLS gRPC ingestion, vectorized DSP anomaly detection, and local LLM triage that falls back to manual review for low-confidence or invalid responses.

AWARDS

Ford Pro Retail and Pro Services Technology Q4 2025 Hackathon Finalist, Oct 2025

EDUCATION

Bachelor of Science in Computer Science, Mar Athanasius College of Engineering, Kothamangalam, Kerala, India; Cumulative GPA: 7.88 / 10